Review of Vector Calculus

Synopsis

A review of key vector calculus topics, including the gradient, Hessian, Jacobian, Taylor's theorem, and Lagrange multipliers.

C.1. THE GRADIENT, HESSIAN, AND JACOBIAN

In vector calculus, the familiar first and second derivatives of a single-variable function are generalized to operate on vectors.

Definition C.1 Given a scalar-valued function with a vector argument, $f(\mathbf{x})$, the **gradient** of f is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}. \tag{C.1}$$

The vector $\nabla f(\mathbf{x})$ has an important geometric interpretation in that it points in the direction in which $f(\mathbf{x})$ increases most rapidly at the point $\mathbf{x}$.

Recall from single-variable calculus that if a function f is continuously differentiable, then a point x^* can only be a minimum or maximum point of f if $f'(x)|_{x=x^*} = 0$. Similarly in vector calculus, if $f(\mathbf{x})$ is continuously differentiable, then a point $\mathbf{x}^*$ can only be a minimum or maximum point if $\nabla f(\mathbf{x}^*) = \mathbf{0}$. In more than one dimension, a point $\mathbf{x}^*$ where $\nabla f(\mathbf{x}^*) = \mathbf{0}$ can also be a **saddle point**. Any point where $\nabla f(\mathbf{x}^*) = \mathbf{0}$ is called a **critical point**.

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Definition C.2 Given a scalar-valued function of a vector, $f(\mathbf{x})$, the **Hessian** of f is a square matrix of partial derivatives given by

$$\mathbf{H}(f(\mathbf{x})) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}. \tag{C.2}$$

If f is twice continuously differentiable, the Hessian is symmetric. It is common in mathematics to write the Hessian using the operator ∇^2, but this sometimes leads to confusion with another vector calculus operator, the Laplacian.

Theorem C.1 *If $f(\mathbf{x})$ is a twice continuously differentiable function, and $\mathbf{H}(f(\mathbf{x}_0))$ is a positive semidefinite matrix, then $f(\mathbf{x})$ is a* **convex function** *at $\mathbf{x}_0$. If $\mathbf{H}(f(\mathbf{x}_0))$ is positive definite, then $f(\mathbf{x})$ is* **strictly convex** *at $\mathbf{x}_0$.*

This theorem can be used to check whether a critical point is a minimum of f. If $\mathbf{x}^*$ is a critical point of f and $\mathbf{H}(f(\mathbf{x}^*))$ is positive definite, then f is convex at $\mathbf{x}^*$, and $\mathbf{x}^*$ is thus a local minimum of f.

It will be necessary to compute derivatives of quadratic forms.

Theorem C.2 *Let $f(\mathbf{x}) = \mathbf{x}^T\mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is an n by n symmetric matrix. Then*

$$\nabla f(\mathbf{x}) = 2\mathbf{A}\mathbf{x} \tag{C.3}$$

and

$$\mathbf{H}(f(\mathbf{x})) = 2\mathbf{A}. \tag{C.4}$$

Definition C.3 Given a vector-valued function of a vector, $\mathbf{F}(\mathbf{x})$, where

$$\mathbf{F}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix}, \tag{C.5}$$

the **Jacobian** of $\mathbf{F}$ is

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}. \tag{C.6}$$

Some authors use the notation $\nabla\mathbf{F}(\mathbf{x})$ for the Jacobian. Note that the rows of $\mathbf{J}(\mathbf{x})$ are the gradients (C.1) of the functions $f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x})$.

C.2. TAYLOR'S THEOREM

In the calculus of single-variable functions, Taylor's theorem produces an infinite series for $f(x+\Delta x)$ in terms of $f(x)$ and its derivatives. Taylor's theorem can be extended to a function of a vector $f(\mathbf{x})$, but in practice, derivatives of order higher than 2 are extremely inconvenient. The following form of Taylor's theorem is often used in optimization theory.

Theorem C.3 *Suppose that $f(\mathbf{x})$ and its first and second partial derivatives are continuous. For any vectors $\mathbf{x}$ and $\Delta\mathbf{x}$, there is a vector $\mathbf{c}$, with $\mathbf{c}$ on the line between $\mathbf{x}$ and $\mathbf{x}+\Delta\mathbf{x}$, such that*

$$f(\mathbf{x}+\Delta\mathbf{x}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T\Delta\mathbf{x} + \frac{1}{2}\Delta\mathbf{x}^T\mathbf{H}(f(\mathbf{c}))\Delta\mathbf{x}. \tag{C.7}$$

This form of **Taylor's theorem with remainder term** is useful in many proofs. However, in computational work there is no way to determine $\mathbf{c}$. For that reason, when $\Delta\mathbf{x}$ is a small perturbation, we often make use of the approximation

$$f(\mathbf{x}+\Delta\mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^T\Delta\mathbf{x} + \frac{1}{2}\Delta\mathbf{x}^T\mathbf{H}(f(\mathbf{x}))\Delta\mathbf{x}. \tag{C.8}$$

An even simpler version of Taylor's theorem, called the **mean value theorem**, uses only the first derivative.

Theorem C.4 *Suppose that $f(\mathbf{x})$ and its first partial derivatives are continuous. For any vectors $\mathbf{x}$ and $\Delta\mathbf{x}$ there is a vector $\mathbf{c}$, with $\mathbf{c}$ on the line between $\mathbf{x}$ and $\mathbf{x}+\Delta\mathbf{x}$ such that*

$$f(\mathbf{x}+\Delta\mathbf{x}) = f(\mathbf{x}) + \nabla f(\mathbf{c})^T\Delta\mathbf{x}. \tag{C.9}$$

We will make use of a truncated version of (C.8):

$$f(\mathbf{x}+\Delta\mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^T\Delta\mathbf{x}. \tag{C.10}$$

By applying (C.10) to each of the functions $f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x})$, we obtain the approximation

$$\mathbf{F}(\mathbf{x}+\Delta\mathbf{x}) \approx \mathbf{F}(\mathbf{x}) + \mathbf{J}(\mathbf{x})\Delta\mathbf{x}. \tag{C.11}$$

C.3. LAGRANGE MULTIPLIERS

The method of **Lagrange multipliers** is an important technique for solving optimization problems of the form:

$$\begin{aligned} \min f(\mathbf{x}) \\ g(\mathbf{x}) = 0, \end{aligned} \tag{C.12}$$

where the scalar-valued function of a vector argument, $f(\mathbf{x})$, is called the **objective function**.

Figure C.1 shows a general situation. The solid contour represents the set of points where the (nonconstant) function $g(\mathbf{x}) = 0$, and the dashed contours are those of another function $f(\mathbf{x})$ that has a minimum as indicated. Moving along the $g(\mathbf{x}) = 0$ contour, we can trace out the curve $\mathbf{x}(t)$, parameterized by the variable $t \geq 0$, where $g(\mathbf{x}(t)) = 0$ and t increases as we progress counter-clockwise. At any point $\mathbf{x}(t)$ on the contour, the gradient of $g(\mathbf{x}(t))$ must be perpendicular to the contour because the function is constant along this curve. Note that in Figure C.1, $g(\mathbf{x})$ increases in the outward direction relative to the contour, so the gradient of $g(\mathbf{x})$ will be outward.

By the chain rule,

$$f'(\mathbf{x}(t)) = \mathbf{x}'(t)^T \nabla f(\mathbf{x}(t)), \tag{C.13}$$

where $\mathbf{x}'(t)$ is the counter-clockwise tangent to the contour $g(x) = 0$. For the point $\mathbf{x}_1 = \mathbf{x}(t_1)$ in Figure C.1, $\nabla f(\mathbf{x}_1)$ and $\mathbf{x}'(t_1)$ are at an obtuse angle, and their dot product $f'(\mathbf{x}_1)$ (C.13) will therefore be negative. Thus, $f(\mathbf{x})$ is decreasing as we move counter-clockwise around the contour $g(\mathbf{x}) = 0$ from $\mathbf{x}_1$, and $\mathbf{x}_1$ cannot satisfy (C.12).

In Figure C.2, for the point $\mathbf{x}_0 = \mathbf{x}(t_0)$, $\nabla f(\mathbf{x}_0)$ is perpendicular to the curve $g(\mathbf{x}) = 0$. In this case, by (C.13), $f'(\mathbf{x}_0) = 0$, and the point $\mathbf{x}_0$ may or may not be a minimum for $f(\mathbf{x})$ along the contour. Figure C.2 shows that a point $\mathbf{x}_0$ on the curve $g(\mathbf{x}) = 0$ can only be a possible minimum point for $f(\mathbf{x})$ if $\nabla g(\mathbf{x}_0)$ and $\nabla f(\mathbf{x}_0)$ are parallel or antiparallel. A point where this occurs is called a **stationary point**.

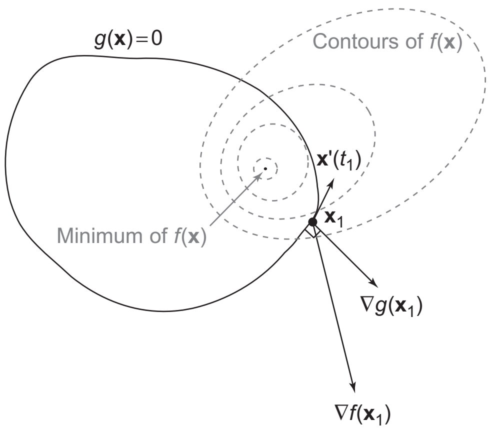

Figure C.1 The situation at a point $\mathbf{x}_1 = \mathbf{x}(t_1)$ along the contour $g(\mathbf{x}) = 0$ that is not a minimum of $f(\mathbf{x})$ and thus does not satisfy (C.12).

Finding a stationary point is necessary, but not sufficient, for finding a minimum of $f(\mathbf{x})$ along the contour $g(\mathbf{x}) = 0$, because such a point may be a minimum, maximum, or saddle point. Furthermore, a problem may have several local minima. Thus it is necessary to examine the behavior of $f(\mathbf{x})$ at all stationary points to find a global minimum.

Theorem C.5 *(C.12) can only be satisfied at a point* $\mathbf{x}_0$ *where*

$$\nabla f(\mathbf{x}_0) + \lambda \nabla g(\mathbf{x}_0) = \mathbf{0} \tag{C.14}$$

for some λ. λ *is called a* **Lagrange multiplier**.

The Lagrange multiplier condition can be extended to problems of the form

$$\begin{aligned} &\min f(\mathbf{x}) \\ &\quad g(\mathbf{x}) \leq 0. \end{aligned} \tag{C.15}$$

Since points along the curve $g(\mathbf{x}) = 0$ are still feasible in (C.15), (C.14) must still hold true. However, there is an additional restriction. Suppose that $\nabla g(\mathbf{x}_0)$ and $\nabla f(\mathbf{x}_0)$ both point in the outward direction, as in Figure C.2. In this case, we can move in the opposite direction, into the feasible region to decrease $f(\mathbf{x})$ (e.g., in the situation depicted in Figure C.2, the solution to (C.15) is simply the indicated minimum of $f(\mathbf{x})$). Thus, a point $\mathbf{x}_0$ satisfying (C.14) cannot satisfy (C.15) unless the gradients of $g(\mathbf{x}_0)$ and $f(\mathbf{x}_0)$ point in opposite directions.

Theorem C.6 *(C.15) can only be satisfied at a point* $\mathbf{x}_0$ *where*

$$\nabla f(\mathbf{x}_0) + \lambda \nabla g(\mathbf{x}_0) = \mathbf{0} \tag{C.16}$$

for some Lagrange multiplier $\lambda > 0$.

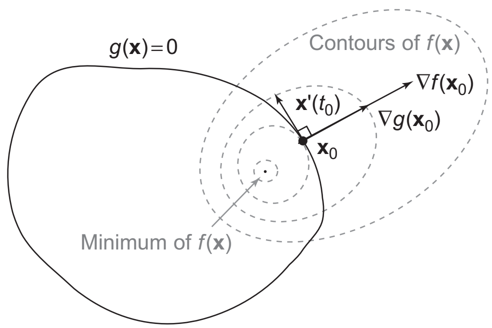

Figure C.2 The situation at a point $\mathbf{x}_0 = \mathbf{x}(t_0)$ along the contour $g(\mathbf{x}) = 0$ that is a minimum of $f(\mathbf{x})$ and thus satisfies (C.12). Note that $\nabla g(\mathbf{x}_0)$ and $\nabla f(\mathbf{x}_0)$ are parallel.

Example C.1

Consider a simple example in two variables where $f(\mathbf{x})$ defines linear contours and $g(\mathbf{x}) = 0$ defines a unit circle

$$\begin{aligned} \min\ & x_1 + x_2 \\ & x_1^2 + x_2^2 - 1 \le 0. \end{aligned} \tag{C.17}$$

The Lagrange multiplier condition is

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix} = \mathbf{0}. \tag{C.18}$$

One stationary point solution to this nonlinear system of equations is $x_1 = 0.7071$, $x_2 = 0.7071$, with $\lambda = -0.7071$. This is the maximum of $f(\mathbf{x})$ subject to $g(\mathbf{x}) \le 0$. The second solution to (C.18) is $x_1 = -0.7071$, $x_2 = -0.7071$, with $\lambda = 0.7071$. Because this is the only solution with $\lambda > 0$, so that $\nabla f(\mathbf{x})$ and $\nabla g(\mathbf{x})$ are antiparallel, this solves the minimization problem.

Note that (C.16) is (except for the non-negativity constraint on λ) the necessary condition for a minimum point of the unconstrained minimization problem,

$$\min f(\mathbf{x}) + \lambda g(\mathbf{x}). \tag{C.19}$$

Here the parameter λ can be adjusted so that, for the optimal solution, $\mathbf{x}^*$, $g(\mathbf{x}^*) \le 0$. We will make frequent use of this technique to convert constrained optimization problems into unconstrained optimization problems.

C.4. EXERCISES

1. Let

$$f(\mathbf{x}) = x_1^2 x_2^2 - 2x_1 x_2^2 + x_2^2 - 3x_1^2 x_2 + 12x_1 x_2 - 12x_2 + 6. \tag{C.20}$$

Find the gradient, $\nabla f(\mathbf{x})$, and Hessian, $\mathbf{H}(f(\mathbf{x}))$. What are the critical points of f? Which of these are minima and maxima of f?

2. Find a Taylor's series approximation for $f(\mathbf{x} + \Delta\mathbf{x})$, where

$$f(\mathbf{x}) = e^{-(x_1+x_2)^2} \tag{C.21}$$

is near the point

$$\mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}. \tag{C.22}$$

3. Use the method of Lagrange multipliers to solve the problem,

$$\begin{aligned} &\min\ 2x_1 + x_2 \\ &4x_1^2 + 3x_2^2 - 5 \leq 0. \end{aligned} \tag{C.23}$$

4. Derive the formula (A.89) for the 2-norm of a matrix. Begin with the maximization problem,

$$\max_{\|\mathbf{x}\|_2=1} \|\mathbf{Ax}\|_2^2. \tag{C.24}$$

Note that we have squared $\|\mathbf{Ax}\|_2$. We will take the square root at the end of the problem.

i. Using the formula $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^T\mathbf{x}}$, rewrite the above maximization problem without norms.
ii. Use the Lagrange multiplier method to find a system of equations that must be satisfied by any stationary point of the maximization problem.
iii. Explain how the eigenvalues and eigenvectors of $\mathbf{A}^T\mathbf{A}$ are related to this system of equations. Express the solution to the maximization problem in terms of the eigenvalues and eigenvectors of $\mathbf{A}^T\mathbf{A}$.
iv. Use this solution to get $\|\mathbf{A}\|_2$.

5. Derive the normal equations (2.3) using vector calculus, by letting

$$f(\mathbf{m}) = \|\mathbf{Gm} - \mathbf{d}\|_2^2 \tag{C.25}$$

and minimizing $f(\mathbf{m})$. Note that in problems with many least squares solutions, all of the least squares solutions will satisfy the normal equations.

i. Rewrite $f(\mathbf{m})$ as a dot product and then expand the expression.
ii. Find $\nabla f(\mathbf{m})$.
iii. Set $\nabla f(\mathbf{m}) = \mathbf{0}$, and obtain the normal equations.

C.5. NOTES AND FURTHER READING

Basic material on vector calculus can be found in many calculus textbooks. However, more advanced topics, such as Taylor's theorem for functions of a vector, are often skipped in basic texts. The material in this chapter is particularly important in optimization, and can often be found in associated references [58, 105, 119].